

TIToK: A solution for bi-imbalanced unsupervised domain adaptation

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ABSTRACT

Unsupervised domain adaptation (UDA) aims to transfer knowledge via domain alignment, and typically assumes balanced data distribution. When deployed in real tasks, however, (i) each domain usually suffers from class imbalance, and (ii) different domains may have different class imbalance ratios. In such bi-imbalanced cases with both within-domain and across-domain imbalance, source knowledge transfer may degenerate the target performance. Some recent efforts have adopted source re-weighting to this issue, in order to align label distributions across domains. However, since target label distribution is unknown, the alignment might be incorrect or even risky. In this paper, we propose an alternative solution named TIToK for bi-imbalanced UDA, by directly *Transferring Imbalance-Tolerant Knowledge* across domains. In TIToK, a class contrastive loss is presented for classification, in order to alleviate the sensitivity to imbalance in knowledge transfer. Meanwhile, knowledge of class correlation is transferred as a supplementary, which is commonly invariant to imbalance. Finally, discriminative feature alignment is developed for a more robust classifier boundary. Experiments over benchmark datasets show that TIToK achieves competitive performance with the state-of-the-arts, and its performance is less sensitive to imbalance.

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1. Introduction

Unsupervised domain adaptation (UDA) (Hoffman et al., 2014; Li, Wang, Shi, Hou, & Liu, 2018; Liang, He, Sun, & Tan, 2019; Sankaranarayanan, Balaji, Jain, Lim, & Chellappa, 2018) transfers knowledge from source domain to help the learning of target domain. Prevalent UDA methodologies mainly attempt to reduce data discrepancy across domains for knowledge transfer (Ganin et al., 2016; Gong, Shi, Sha, & Grauman, 2012; Long, Cao, Wang, & Jordan, 2015; Long, Zhu, Wang, & Jordan, 2016, 2017; Pan, Tsang, Kwok, & Yang, 2010; Pei, Cao, Long, & Wang, 2018; Tzeng, Hoffman, Saenko, & Darrell, 2017; Zhang, Ouyang, Li, & Xu, 2018). Specifically, the marginal or conditional feature distributions are aligned by reducing some discrepancy metric (Long, Cao, Cao, Wang, & Jordan, 2018a), or by adversarial learning between feature generator and domain discriminator (Ganin et al., 2016). After that, the classifier learned from labeled source domain is applied to target instances. As a result, current UDA methods mainly focus on feature distribution shift, i.e., $p_s(\mathbf{x}) \neq p_t(\mathbf{x})$

or $p_s(\mathbf{x}|\mathbf{y}) \neq p_t(\mathbf{x}|\mathbf{y})$, and assume that classes are relatively balanced within each domain, and the balance ratio is similar across domains.

However, raw data in real-world scenarios are usually imbalanced. The imbalance often occurs within each domain, and it may further induce different class proportions or imbalance ratios across domains, termed across-domain imbalance. For example, in automatic driving, the perceived environment is constantly changing, *automobiles* are much more frequently seen in urban than rural area, while *cattle* are more common in rural area. UDA learning usually encounters such bi-imbalance scenario with imbalance within each domain and across different domains, i.e., $p_{s/t}(\mathbf{y}_i) \neq p_{s/t}(\mathbf{y}_j)$, $\forall i \neq j$ and $p_s(\mathbf{y}) \neq p_t(\mathbf{y})$. In such cases, the majority source class can be exactly a minority class in target domain. Previous UDA methods, which only consider feature discrepancy across domains, may degenerate the target performance with negative transfer (Zhao, Des Combes, Zhang, & Gordon, 2019). As an illustration shown in Fig. 1, without considering the bi-imbalance issue, classification boundary will bias toward majority classes in source domain, which are unfortunately minority classes in target domain. It consequently leads to misclassification of majority target instances, or even negative transfer.

Recently, some efforts have adopted source re-weighting to this issue (Cheng, Guo, Candan, & Liu, 2020; Jiang, Lao, Matwin, & Havaei, 2020). However, instance re-weighting has the risk of

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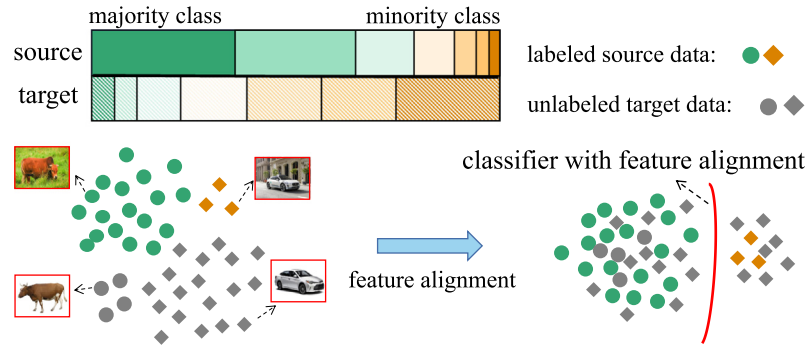


Fig. 1. The learning scenario of bi-imbalanced UDA, and classifier after feature alignment.

knowledge loss from majority classes after down-weighting, or overfitting for minority classes after over-weighting (Kim, Jeong, & Shin, 2020). Moreover, they commonly rely on label distribution alignment across domains, which might be incorrect or even risky with target label distribution unknown. In this paper, we propose an alternative way for bi-imbalanced UDA termed TIToK, by directly transferring the imbalance-tolerant knowledge across domains. Specifically, inspired by the imbalance-insensitive metric AUC (area under the ROC curve) (Yang, Xu, Bao, Cao, & Huang, 2021), a class contrastive loss is presented for classification in UDA, in order to alleviate the sensitivity to class imbalance in knowledge transfer, and it is adopted over both source and unlabeled target data. Meanwhile, class correlation knowledge from source domain is transferred as a supplementary via class correlation transfer, which is commonly invariant to imbalance. Finally, discriminative feature alignment is developed with both intra-class alignment and inter-class separation, for a more robust classifier boundary. Experiments on various benchmark datasets consistently show that TIToK achieves competitive performance compared with state-of-the-arts, and its performance is less sensitive to class imbalance.

The contributions of this paper are as follows: (i) An alternative way for bi-imbalanced UDA is proposed by transferring the imbalance-tolerant knowledge across domains. (ii) A systemic solution, including class contrastive knowledge transfer, class correlation knowledge transfer and discriminative feature alignment, is proposed for bi-imbalanced UDA. (iii) State-of-art results for TIToK are reported in bi-imbalanced UDA tasks.

The rest of the paper is organized as follows, Section 2 introduces some related work, Section 3 describes the proposed TIToK method, Section 4 shows the empirical results compared with the state-of-the-arts methods, and some conclusions are given in Section 5.

2. Related work

2.1. Unsupervised domain adaptation

Deep UDA methods achieve the state-of-art performance in UDA learning, and can mainly be divided into three categories: discrepancy-based methods, adversarial methods and self-training methods. Discrepancy-based UDA methods attempt to reduce the domain discrepancy by minimizing some distance metric, such as maximum mean discrepancy (MMD) (Long et al., 2015), correlation (CORAL) (Sun & Saenko, 2016), Kullback–Leibler (KL) divergence (Zhuang, Cheng, Luo, Pan, & He, 2015) and Wasserstein Distance (Lee, Batra, Baig, & Ulbricht, 2019). For example, Tzeng et al. introduce an adaptation layer, as well as an domain confusion loss based on MMD, to learn domain invariant representation (Tzeng, Hoffman, Zhang, Saenko, & Darrell,

2014). Ghifary et al. adopt three adaptation layers, and multi-kernel MMD for domain adaptation (Ghifary, Kleijn, & Zhang, 2014). Lee et al. introduce Sliced Wasserstein Distance (SWD) (Lee et al., 2019), and Kang et al. leverage CAN (Contrastive Domain Discrepancy) (Kang, Jiang, Yang, & Hauptmann, 2019) for UDA. Adversarial UDA methods commonly align features across domains through adversarial learning. An adversarial UDA method is proposed in Ganin et al. (2016), which adopts a reverse gradient layer (RevGrad) to achieve adversarial domain alignment. Tzeng et al. adopt a similar adversarial strategy but a separate feature extractor for each domain (Tzeng et al., 2017). Pei et al. adopt class-wise domain discriminators to enable fine-grained alignment of data distributions (Pei et al., 2018). Motivated by Conditional GANs (Mirza & Osindero, 2014), Jin et al. propose CDAN (Long, Cao, Wang, & Jordan, 2018b) to align domain features in a class-conditional adversarial learning manner. Saito, Watanabe, Ushiku, and Harada (2018a) propose MCD (Maximum Classifier Discrepancy) with adversarial learning across two classifiers. Yang, Xia, Ding, and Ding (2020) propose bi-directional generation with adversarial learning, in order to interpolate intermediate domains between source and target. Researches also adopt self-training strategy for UDA. Zou et al. propose a confidence regularized self-training (CRST) framework for UDA, in which pseudo-labels are treated as continuous latent variables optimized via alternating optimization (Zou, Yu, Liu, Kumar, & Wang, 2019). Kumar et al. adopt gradual self-training to adapt an initial source classifier towards the target domain gradually (Kumar, Ma, & Liang, 2020). Liu et al. propose a cycle self-training (CST) method for UDA with a forward step and reverse step until convergence (Liu, Wang, & Long, 2021).

2.2. Learning with label distribution shift

The main strategies for class imbalanced learning include data re-sampling and cost-sensitive learning. Data re-sampling methods attempt to balance the distribution by either down-sampling the majority classes (Drummond, 2003) or over-sampling the minority classes (Chawla, Bowyer, Hall, & Kegelmeyer, 2002). However, those methods may have the risk of losing important knowledge of the majority classes after down-sampling, or overfitting minority classes after over-sampling (Kim et al., 2020). Cost-sensitive learning commonly assign higher weights to loss of minority classes, in order to penalize more on making mistakes for the minority classes. Another important line directly optimizes a balanced performance metric in learning, such as the area under the ROC curve (AUC).

There are also researches focusing on label distribution shift across domains. Lipton et al. propose Black-Box Shift Learning (BBSL) by model retraining with importance weights (Lipton, Wang, & Smola, 2018). Azizzadenesheli et al. further improve BBSL with regularization items (Azizzadenesheli, Liu, Yang, &

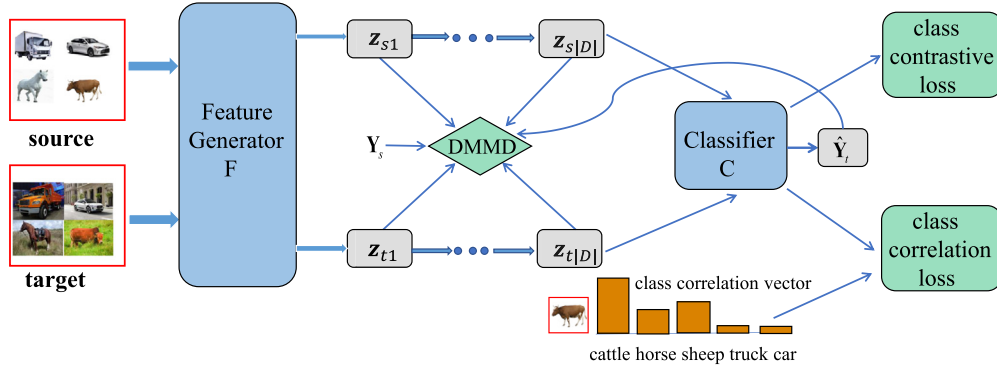


Fig. 2. Network structure of TIToK. Class contrastive loss is adopted over both source and confident target instances for imbalance-tolerant knowledge transfer, class correlation loss is adopted to preserve class correlation knowledge from source domain, DMMD is minimized for discriminative distribution alignment across domains.

Anandkumar, 2018). Amr et al. propose to combine maximum likelihood with bias-corrected calibration (Alexandari, Kundaje, & Shrikumar, 2020). Kamyar et al. propose a series of estimators for importance weights from source to target domain (Azzizadehsheli, 2021). However, they commonly assume that the feature distribution is invariant across domains.

Recently, researchers adopt instance re-weighting to address label distribution shift in UDA, in order to align both feature and label distributions across domains. Hsu, Wei, Hou, Tsai, and Wang (2015) learn latent source domain by instance weights relating within and cross-domain data pairs in adaptation. A weighted MMD is proposed for UDA (Yan et al., 2017), which assigns weight for each source class using EM strategy. Jing, Xu, and Ding (2021) adopt cross-domain mixup generation for minority source sets. Cheng et al. (2020) jointly learn re-weighting strategy and representation to simultaneously account for covariate and representation biases. Tan, Peng, and Saenko (2020) address both data and label distribution shift by combining self-training and conditional feature alignment. Jiang et al. (2020) adopt instance sampling in MDD alignment guided by pseudo-labels, while Prabhu, Khare, Kartik, and Hoffman (2021) also adopt pseudo-label-based sampling in UDA of self-training manner. Liu, Guo, et al. (2021) propose to estimate the target label distribution using marginal matching. Though the learning scenario is similar to ours, those methods commonly adopt re-weighting strategies, which may cause knowledge loss or overfitting issue described above. Moreover, they commonly rely on estimation of target pseudo-labels for label distribution alignment, which might be incorrect or even risky with error accumulation. To this end, we attempt to propose an alternative way for bi-imbalance UDA, by directly transferring imbalance-tolerant knowledge across domains.

3. TIToK: transferring imbalance-tolerant knowledge

In this section, in order to address both feature and label distribution shift, we propose an alternatively way by transferring imbalance-tolerant knowledge across domains. Specifically, we propose a systemic solution to deal with bi-imbalance UDA, mainly containing three components, i.e., *class contrastive knowledge transfer*, *class correlation knowledge transfer* and *discriminative feature alignment*.

3.1. Notations and network structure

UDA aims to transfer knowledge from a related source domain to learn accurate classifier for target domain, and the feature distributions are commonly different across domains in UDA. For K -class classification problem, given labeled source instances

$(\mathbf{X}_s, \mathbf{Y}_s) = \{(\mathbf{x}_1^s, \mathbf{y}_1^s), \dots, (\mathbf{x}_{n_s}^s, \mathbf{y}_{n_s}^s)\} \in D_s$ drawn from $p_s(\mathbf{x}, \mathbf{y})$, and unlabeled target instances $\mathbf{X}_t = \{\mathbf{x}_1^t, \dots, \mathbf{x}_{n_t}^t\} \in D_t$ drawn from $p_t(\mathbf{x})$, each $\mathbf{x}_i \in \mathbb{R}^d$, $\mathbf{y}_i \in \mathbb{R}^K$, the bi-imbalance UDA is defined as follows,

Definition 1 (Bi-imbalance UDA). Given labeled source data $(\mathbf{X}_s, \mathbf{Y}_s)$ and unlabeled target data $\mathbf{X}_t$, feature distributions across domains are different, i.e., $p_s(\mathbf{x}) \neq p_t(\mathbf{x})$ or $p_s(\mathbf{x}|\mathbf{y}) \neq p_t(\mathbf{x}|\mathbf{y})$. Meanwhile, there is also within-domain and across-domain imbalance in adaptation, i.e., $p_{s/t}(\mathbf{y}_i) \neq p_{s/t}(\mathbf{y}_j)$, $\forall i \neq j$ and $p_s(\mathbf{y}) \neq p_t(\mathbf{y})$. Bi-imbalance UDA aims to seek an accurate classifier $h: \mathbb{R}^d \rightarrow \mathbb{R}^K$ for target data with the help of source data.

We propose TIToK to solve bi-imbalance UDA, and its main structure is given in Fig. 2. Both source and target instances are sent into a common feature generator F , and the feature distributions are aligned by reducing discriminative MMD (DMMD) for both intra-class alignment and inter-class separation across domains. The newly-generated source and target features are denoted by $\mathbf{z}_s = \{\mathbf{z}_{s1} \dots \mathbf{z}_{s|D|}\}$ and $\mathbf{z}_t = \{\mathbf{z}_{t1} \dots \mathbf{z}_{t|D|}\}$, respectively, where $|D|$ is the dimension of new features. Those features are then sent into a classifier C with K -dimensional probabilistic outputs of classification scores, each element represents the probability of an instance to each class. The class contrastive loss is adopted to constrain that classification scores assigned for instances inside a specific class should be higher than those of instances outside the class. Moreover, correlation among source classes is also transferred via class correlation knowledge transfer. The correlation knowledge represented by class correlation vector is first learned from source domain, and then adapted to target domain with class correlation loss.

3.2. Class contrastive knowledge transfer

We propose to transfer imbalance-tolerant knowledge for bi-imbalance UDA. The cross-entropy loss commonly used in previous UDA methods focuses on assigning source instances to the correct classes, it may lead to class bias to majority classes under bi-imbalance condition, and thus it is sensitive to imbalance. In recent works, source re-weighting is commonly adopted to address the class imbalance. While inspired by the imbalance-insensitive metric, we propose a class contrastive loss for class contrastive knowledge transfer, in order to alleviate the sensitivity to imbalance in adaptation.

AUC is the area under the Receiver Operating Characteristic (ROC) curve, which is plotted by sensitivity against specificity. Equivalent to the Wilcoxon–Mann–Whitney statistic, it works by ranking positive/negative instances in terms of the classification

score, which naturally avoids the dependency on label distribution priors. As a result, it is well-known to be insensitive to class imbalance (Yang et al., 2021). Inspired by AUC, we propose class contrastive loss for bi-imbalanced UDA. Specifically, for multi-class learning, we adopt a single classifier with K -dimensional probabilistic outputs here, each describing the probability of an instance belonging to each class. For the k -th class, we first define the class contrastive label as

$$y_{ij}^k = \begin{cases} 1, & \text{if } \mathbf{x}_i \in \mathbf{X}_k, \mathbf{x}_j \notin \mathbf{X}_k \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

Let $f_k(\mathbf{x}_i)$ denote the classification score of $\mathbf{x}_i$ to the k -th class, we propose that if $\mathbf{x}_i \in \mathbf{X}_k$ and $\mathbf{x}_j \notin \mathbf{X}_k$, then $f_k(\mathbf{x}_i)$ should be higher than $f_k(\mathbf{x}_j)$. Finally the empirical risk with class contrastive loss can be formulated as

$$\mathcal{L}_{con}(\mathbf{X}, \mathbf{Y}; \theta_F, \theta_C) = \sum_{k=1 \dots K} \sum_{\mathbf{x}_i \in \mathbf{X}_k} \sum_{\mathbf{x}_j \notin \mathbf{X}_k} \mathcal{L}(f_k(\mathbf{x}_i) - f_k(\mathbf{x}_j), y_{i,j}^k) \quad (2)$$

where $\mathcal{L}(\cdot)$ is the loss function, θ_F and θ_C are parameters of feature generator and classifier, respectively. The risk is defined over all instance pairs $(\mathbf{x}_i, \mathbf{x}_j)$ satisfying $\mathbf{x}_i \in \mathbf{X}_k$ and $\mathbf{x}_j \notin \mathbf{X}_k$. Minimizing $\mathcal{L}_{con}$ aims to maximize the prediction difference between instances belonging and not belonging to the k -th category. After applying the class contrastive loss, class contrastive knowledge from source domain will be transferred. It actually constrains the ranking of classification outputs between different classes, regardless of the classification threshold between categories. As a result, it does not depend on the proportion among classes, or it is insensitive to class imbalance. With class contrastive knowledge transferred, it is not necessary to align label distributions across domains for bi-imbalanced learning, since the source knowledge transferred involves no class basis. The exponential loss (Yang et al., 2021) is adopted here, then empirical risk in Eq. (2) can be further formulated as

$$\mathcal{L}_{con}(\mathbf{X}, \mathbf{Y}; \theta_F, \theta_C) = \sum_{k=1 \dots K} \sum_{\mathbf{x}_i \in \mathbf{X}_k} \sum_{\mathbf{x}_j \notin \mathbf{X}_k} \exp(-\beta(f_k(\mathbf{x}_i) - f_k(\mathbf{x}_j))) \quad (3)$$

where β is a user-defined parameter. Since the class contrastive loss involves the comparison between each instance pair, the complexity achieves $O(\sum_{k=1 \dots K} \sum_{l \neq k} n_k n_l T)$, where T is the base complexity for each pair, n_k and n_l denote the number of instances in class k and l , respectively. To accelerate the computation, the class contrastive loss is further simplified by a factorization scheme,

$$\mathcal{L}_{con}(\mathbf{X}, \mathbf{Y}; \theta_F, \theta_C) = \sum_{k=1 \dots K} \underbrace{\left(\sum_{\mathbf{x}_i \in \mathbf{X}_k} \exp(-\beta f_k(\mathbf{x}_i)) \right)}_{(a_i)} \cdot \underbrace{\left(\sum_{\mathbf{x}_j \notin \mathbf{X}_k} \exp(\beta f_k(\mathbf{x}_j)) \right)}_{(b_j)} \quad (4)$$

From the derivation above, the loss evaluation can be done by first calculating (a_i) and (b_j) separately, and then performing the multiplication, which only takes a complexity of $O(NT)$, where N is the total number of training instances.

Finally, the class contrastive loss is applied over both source and confident target instances, that is

$$\mathcal{L}_{cc} = \mathcal{L}_{con}(\mathbf{X}_s, \mathbf{Y}_s; \theta_F, \theta_C) + \mathcal{L}_{con}(\mathbf{X}_t^c, \hat{\mathbf{Y}}_t^c; \theta_F, \theta_C) \quad (5)$$

where $\mathbf{X}_t^c$ are confident target instances selected with the maximum classification score larger than a threshold τ , and $\hat{\mathbf{Y}}_t^c$ are the predicted labels.

It is noted that though we abuse the contrastive loss here, our class contrastive loss differs from the contrastive loss in self-supervised learning. Since class contrastive loss in TIToK considers the ranking of classification outputs between instance inside and outside a specific class, in order to address the class imbalance issue. While self-supervised contrastive learning concentrates more on whether two instances belonging to the same class. Moreover, TIToK differs from Contrastive Adaptation Network (CAN) (Kang et al., 2019) in that CAN still adopts CE loss for classification, and proposes Contrastive Domain Discrepancy (CDD) to replace MMD, which is established on the difference between conditional data distributions across domains.

3.3. Class correlation knowledge transfer

The conceptual knowledge of class correlation is commonly invariant across domains (Tzeng, Hoffman, Darrell, & Saenko, 2015), and insensitive to imbalance as well, e.g., category *cattle* is similar to *horse* but different from *automobiles*. As a result, it is transferred across domains in TIToK to further boost knowledge transfer. Class contrastive knowledge involves comparison between classes, while class correlation knowledge preserves correlation structure amongst all categories as a supplementary.

In order to catch the class correlation of source domain, a source correlation vector $\mathbf{c}^{(k)}$ is defined for each category. It is represented by the average softmax outputs for instances within that category. Denote $f(\mathbf{x}_i)$ as $\mathbf{p}_i$, which is the probabilistic classification output for instance $\mathbf{x}_i$, then $\mathbf{c}^{(k)} = \sum_{\mathbf{x}_i \in \mathbf{X}_k^s} \mathbf{p}_i^s$. The correlation vector is first computed over source data, and actually reflects the correlation of each category to the others, e.g., if $c_l^{(k)} > c_m^{(k)}, \forall l, m \neq k$, then category k is more similar to category l than m .

For confident target instances with predicted pseudo-labels, we enforce the consistency between the classifier outputs and the corresponding correlation vector with

$$\mathcal{L}_{cor} = - \sum_{\mathbf{x}_i \in \mathbf{X}_t} c_i^{(\hat{y}_i)} \log \mathbf{p}_i \quad (6)$$

where $c_i^{(\hat{y}_i)}$ is the correlation vector of class that instance $\mathbf{x}_i$ belongs to. In this way, class correlation among categories in source domain is transferred to target domain, in order to enforce a label correlation adaptation across domains.

3.4. Discriminative feature alignment

In the paper, we adopt MMD for feature alignment as an example, in order to show the effectiveness of imbalance-insensitive knowledge transfer across domains. Moreover, it has also been theoretically proved that the learning bound in imbalanced learning involves the margin for each class (Cao, Wei, Gaidon, Arechiga, & Ma, 2019). As a result, we propose discriminative MMD (DMMD) to align intra-class distributions, and separate inter-class distributions across domains at the same time, in order to enlarge the margin among classes.

Further, by introducing soft labels, DMMD across domains is formulated as

$$\mathcal{L}_{DMMD} = \frac{1}{K} \sum_{k=1 \dots K} \left\| \sum_{\mathbf{x}_i^s \in \mathbf{X}_s} u_i^{sk} \phi(\mathbf{x}_i^s) - \sum_{\mathbf{x}_j^t \in \mathbf{X}_t} u_j^{tk} \phi(\mathbf{x}_j^t) \right\|^2 - \left\| \sum_{\mathbf{x}_i^s \in \mathbf{X}_s} \phi(\mathbf{x}_i^s) - \sum_{\mathbf{x}_j^t \in \mathbf{X}_t} \phi(\mathbf{x}_j^t) \right\|^2 \quad (7)$$

Algorithm 1: TIToK

Input: Labeled source instances ($\mathbf{X}_s, \mathbf{Y}_s$),
unlabeled target instances ($\mathbf{X}_t$),
maximum number of epochs E , number of batches B ,
parameters λ and γ .
Output: Optimal parameters θ_F, θ_C of feature generator F and
classifier C .
Initialize θ_F using pre-trained model, and randomly initialize
 θ_C .
Calculate correlation vectors $\mathbf{c}^{(k)}$ for source classes.
for epoch = 1 **to** E **do**
 for iteration = 1 **to** B **do**
 Sample a mini-batch of source and target instances.
 Predict pseudo-labels for target instances:
 $\hat{\mathbf{y}}_t = \text{argmax}_{\theta_C}(\mathbf{y}|\mathbf{x}_t)$.
 Calculate all losses from Eq. (5), (6) and (7).
 Update θ_F and θ_C by minimizing Eq. (8).
 end
 Update pseudo-labels of target instances $\hat{\mathbf{Y}}_t$.
end

where weight u_i^{sk} denotes the probability of source instance $\mathbf{x}_i^s$ to the k -th class, u_j^{tk} is the probability of target instance $\mathbf{x}_j^t$ to the k -th class. The u_i^{sk} can be directly obtained from source labels, i.e., u_i^{sk} equals 1 if $\mathbf{x}_i^s$ belongs to the k -th class, and 0 otherwise. u_j^{tk} s are set to the probabilistic outputs of classifier, i.e., $u_j^{tk} = f_k(\mathbf{x}_j^t)$. Intuitively, the first and second items in Eq. (7) are the intra-class and global discrepancy across domains, respectively. Minimizing Eq. (7) actually enlarges the overall distance across domains, while simultaneously reduces the intra-class discrepancy. As a result, it actually enforces the intra-class feature alignment, and separates the inter-class distributions at the same time.

Previous work of CAN also considers to minimize the intra-class domain discrepancy and maximize the inter-class domain discrepancy with CDD (Kang et al., 2019). The inter-class domain discrepancy in CDD is formulated by sum of discrepancy between each category pair. While DMMD concentrate on minimizing the intra-class compactness and maximizing the global separation at the same time, in order to enforce intra-class feature alignment and inter-class feature discrepancy.

3.5. Overall objective and optimization

TIToK considers the class contrastive knowledge transfer, class correlation knowledge transfer and discriminative feature alignment at the same time, and seeks the parameters for feature generator F and classifier C by minimizing the following joint objective,

$$\min_{\theta_F, \theta_C} \mathcal{L}_{cc} + \lambda \mathcal{L}_{DMMD} + \gamma \mathcal{L}_{cor} \quad (8)$$

where λ and γ are hyper-parameters to balance the contributions of different items. Given input instances from both source and target domains, networks F and C are first pretrained with only source instances by minimizing $\mathcal{L}_{con}(\mathbf{X}_s, \mathbf{Y}_s; \theta_F, \theta_C)$. Then pseudo-labels of target instances are assigned, and the class correlation vectors for source domain are computed, with which the networks are updated by minimizing Eq. (8). With imbalance-tolerant knowledge transferred, target pseudo-labels are more accurate under bi-imbalanced condition. Finally, by iteration between pseudo-label assignment and network learning, better feature representation and more accurate pseudo-labels of target instances can be obtained. The algorithm description of TIToK is shown in Algorithm 1.

4. Experiments

In this section, we compare the performance of TIToK with the state-of-art methods over both object recognition and digit classification tasks. Two datasets, Office-Home and Domain-Net, are used for object recognition task. For digit classification, the handwritten digits dataset is adopted, and the transfer tasks are constructed from MNIST, USPS, and SVHN. All transfer learning tasks are denoted as source domain $\rightarrow$ target domain. We first give the experimental setups, then show the results of performance comparison. We also provide the method analysis, including ablation study, sensitivity to parameters, and feature visualization as well. The code of TIToK is available at <https://github.com/bigchuan-bro/TiTok>.

4.1. Experiment setups

Office-Home is a benchmark dataset for domain adaptation collected in office and home environment. It includes 4 domains: Real World (Rw), Product (Pr), Clipart (Cl) and Art (Ar). Each domain contains 65 classes and 15588 images. Since the Art domain is too small for an imbalanced subset, we focus on the rest 3 domains and explore 6 transfer tasks, $Rw \rightarrow Pr$, $Rw \rightarrow Cl$, $Pr \rightarrow Rw$, $Pr \rightarrow Cl$, $Cl \rightarrow Rw$ and $Cl \rightarrow Pr$.

DomainNet is a large-scale dataset for domain adaptation, which contains six domains with about 0.6 million images distributed over 345 classes. Since some domains and classes contain lots of mislabeled outliers, we select 40 commonly-used classes from 3 domains: Real (R), Clipart (C), Painting (P), thus six transfer tasks, $R \rightarrow C$, $R \rightarrow P$, $C \rightarrow R$, $C \rightarrow P$, $P \rightarrow R$ and $P \rightarrow C$.

Digits is a handwritten digits dataset containing three domains, i.e., SVHN (S) with 73,257 images, USPS (U) with 7,291 images and MNIST with 60,000 images. Each domain has 10 classes. Finally, experiments are conducted over 3 transfer tasks, $USPS \rightarrow MNIST$, $MNIST \rightarrow USPS$ and $SVHN \rightarrow MNIST$.

Sample images of the three datasets are shown in Fig. 3.

4.1.1. Baseline methods

TIToK is compared with both UDA methods based on discrepancy minimization and adversarial learning, respectively, as well as recent UDA methods for imbalance learning. The compared methods are listed below,

Source only is the model trained with only source samples.

DAN (Deep adaptation network) (Long et al., 2015) and *JAN* (Joint Adaptation Network) (Long et al., 2017) are discrepancy-based deep UDA methods with marginal and joint distribution alignment, respectively.

DANN (Domain adversarial neural network) (Ganin et al., 2016) and *MCD* (Maximum Classifier Discrepancy) (Saito, Watanabe, Ushiku, & Harada, 2018b) both adopt adversarial strategy to align feature distribution across domains. *BSP* (Batch Spectral Penalization) (Chen, Wang, Long, & Wang, 2019) further enhances the discriminability of adversarial models by penalizing the largest singular values.

FDANN (Admissible f -divergence DANN) (Wu, Winston, Kaushik, & Lipton, 2019) relaxes the feature distribution alignment in *DANN* for dealing with potential label shift. *BBSE* (Black Box Shift Estimation) (Lipton et al., 2018) only aligns label distributions across domains. *COAL* (CO-Alignment) (Tan et al., 2020) addresses data imbalance and label distribution shift simultaneously by combining self-training and conditional feature alignment.

Table 1
Per-class average accuracy (%) on Office-Home.

Methods	Rw→Pr	Rw→Cl	Pr→Rw	Pr→Cl	Cl→Rw	Cl→Pr	AVG
Source only	69.71	38.35	67.31	35.84	53.31	52.27	52.81
DAN	67.85	38.17	66.86	34.24	53.55	51.64	52.05
JAN	67.20	43.60	68.87	39.21	57.98	48.57	54.24
DANN	71.62	46.51	67.98	38.07	58.83	57.39	56.73
MCD	66.03	33.17	62.95	29.99	44.47	39.01	45.94
BSP	72.80	23.82	66.19	20.81	34.54	31.04	41.53
BBSE	61.10	33.27	62.66	31.15	39.70	38.08	44.33
F-DANN	68.56	40.57	67.32	37.33	55.84	53.67	53.88
COAL	73.65	42.58	73.26	40.61	59.22	62.71	58.67
TIToK	77.09	52.84	72.15	44.32	60.06	59.95	61.07

Table 2
Per-class average accuracy (%) on DomainNet.

Methods	R→C	R→P	C→R	C→P	P→R	P→C	AVG
Source only	58.84	67.89	76.70	53.55	84.31	55.55	66.14
DAN	64.36	70.65	79.44	56.78	84.56	61.62	69.57
JAN	65.57	73.58	85.02	64.96	87.06	67.92	74.02
DANN	63.37	73.56	86.47	65.73	86.94	73.19	74.88
MCD	61.97	69.33	79.78	56.61	83.38	58.31	68.23
BSP	67.29	73.47	86.50	67.52	86.83	70.33	75.32
BBSE	55.38	63.62	64.58	42.18	81.55	49.04	59.39
F-DANN	66.15	71.80	81.85	60.06	84.46	66.81	71.86
COAL	73.85	75.37	89.63	69.98	89.81	68.01	77.78
TIToK	77.04	75.48	87.58	69.34	88.35	74.44	78.71

Table 3
Per-class average accuracy (%) on Digits.

Methods	U→M	M→U	S→M	AVG
Source Only	75.31	87.92	50.25	71.16
DAN	79.12	87.15	53.63	73.3
JAN	75.75	78.82	53.21	69.26
DANN	77.28	91.88	57.16	75.44
MCD	77.18	85.34	53.52	72.01
BSP	71.99	89.74	50.61	70.78
F-DANN	72.59	81.62	45.65	66.62
BBSE	75.01	78.84	49.01	67.62
COAL	88.12	93.04	65.67	82.28
TIToK	88.96	94.61	67.28	83.62

Table 4
Ablation study for TIToK.

Methods	Rw→Pr	Rw→Cl	R→C	P→C
baseline	67.85	39.21	64.36	61.63
$\mathcal{L}_{DMMD}$	70.45	46.10	72.24	69.43
$\mathcal{L}_{DMMD} + \mathcal{L}_{cc}$	76.12	49.34	76.08	72.92
$\mathcal{L}_{DMMD} + \mathcal{L}_{cc} + \mathcal{L}_{cor}$	77.09	52.84	77.04	74.44

Office-Home and DomainNet datasets. We also use DAN as the baseline, and the results are shown in Table 4.

From those results, we can find that the three components in TIToK all help boost the target learning under the bi-imbalanced learning scenario. For example, for $Rw \rightarrow Cl$, through minimizing DMMD instead of MMD, the per-class average accuracy is improved by 6.89%, when class contrastive transfer is added, the performance increases by 3.24%. Further, if class correlation knowledge transfer is introduced, the performance increases by 3.5%. Those results demonstrate that the components of TIToK are all helpful for bi-imbalance UDA learning.

4.3.2. Tolerance to imbalance

In Fig. 4, we reveal the per-class target accuracies with class contrastive loss in both balanced and bi-imbalanced scenarios over USPS→MNIST, in order to show its tolerant to imbalance.

Specifically, Fig. 4(a) and (b) show the performance with cross-entropy loss in balanced and bi-imbalanced cases, respectively, while (c) and (d) show the performance with class contrastive loss in balanced and bi-imbalanced cases, respectively. From Fig. 4, with cross-entropy loss, the per-class target accuracies span a wider range in bi-imbalanced case than in balanced case. It leads to poor performance for majority target classes, which are minority classes in source domain. While UDA with proposed class contrastive loss can obtain more unbiased per-class target accuracies, as well as better average accuracy, demonstrating that transferring class contrastive knowledge with class contrastive loss is more tolerant to the imbalance.

4.3.3. Different degrees of imbalance

In previous experiments, performance comparison is conducted under a certain imbalance condition. In this sub-section, we further analysis the performance under different degrees of imbalance over the USPS → MNIST task, and the results are shown in Fig. 5. In Fig. 5, “0.0” means that both source and target domains are balanced, while “1.0” represents that the class distribution from source domain is exactly a reversed version of that from target domain. Afterwards we create 4 interval degrees between 0.0 and 1.0 using linear interpolation. The proportions of each category is computed with parameter β from 0.2 to 0.8. For example, the proportion of class “1” with current β is computed by the proportions of class “1” in cases “0.0” and “1”, respectively, denoted by p_1 and p_2 , i.e., $p = (p_1 + \beta * (p_2 - p_1))$. In each case, we randomly select instances from each class according to the proportions obtained.

From Fig. 5, it can be seen that performance of previous UDA methods is significantly affected by the degree of imbalance. As the imbalance degree increases, performances of both MCD and DANN drop rapidly, and the performance of MCD drops faster than that of DANN. At the same time, TIToK and COAL are relatively stable compared with the other methods, since they both consider the imbalance issue in learning. Moreover, TIToK is more stable and achieves better performance, demonstrating the robustness of TIToK for bi-imbalanced UDA.

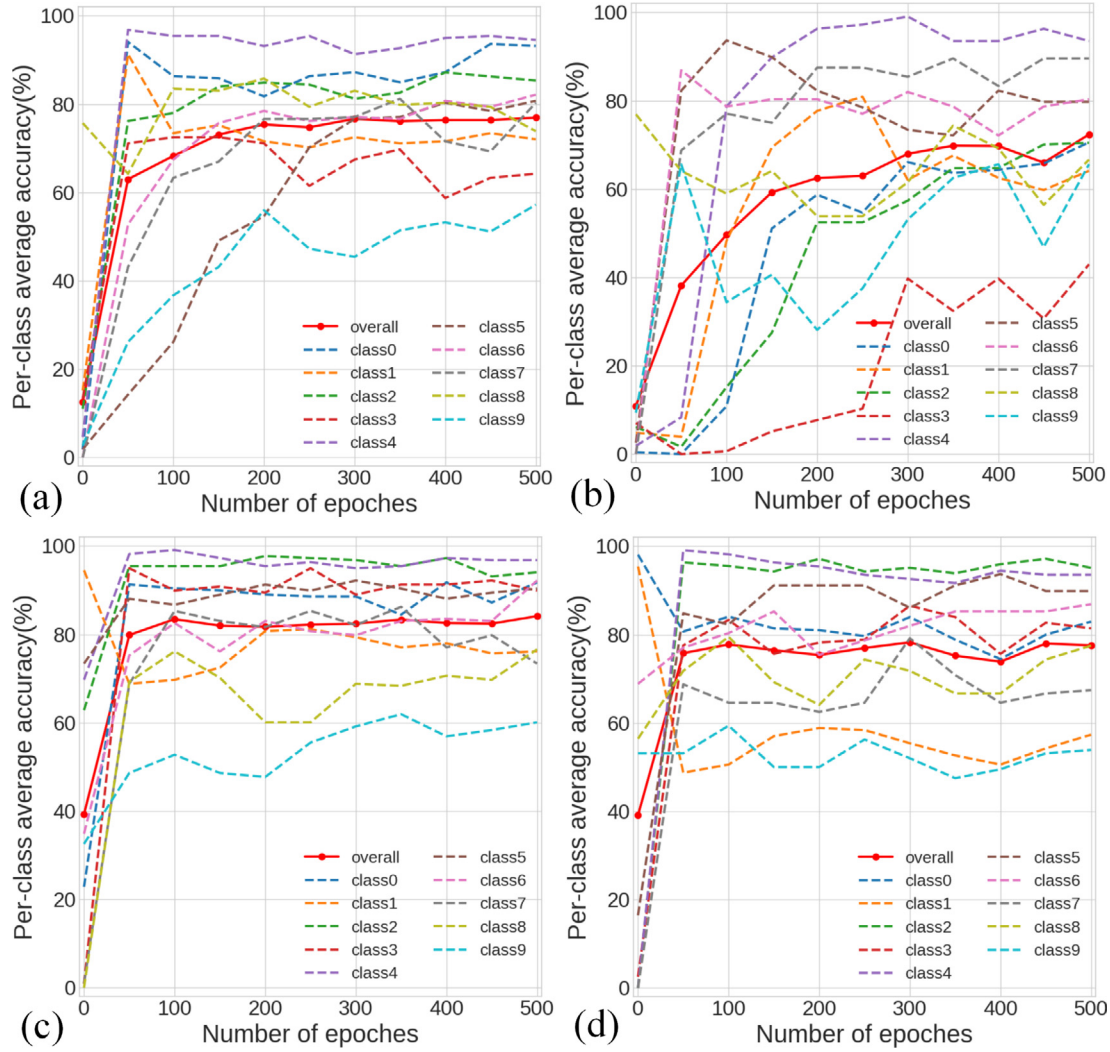


Fig. 4. Per-class and overall target accuracies with cross-entropy loss in (a) balanced and (b) bi-imbalanced scenarios, and with the proposed class contrastive loss in (c) balanced and (d) bi-imbalanced scenarios. In the bi-imbalanced case, classes 0~3 are minority classes in source domain, but majority classes in target domain.

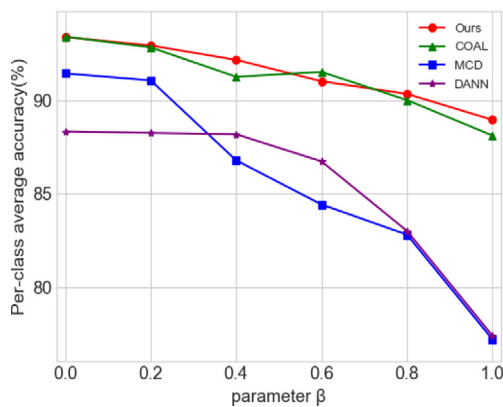


Fig. 5. Performance on USPS→MNIST task with different imbalance degrees.

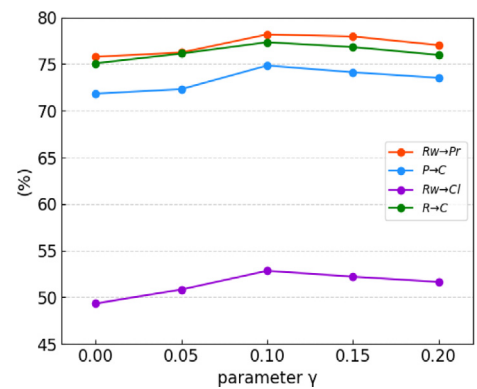


Fig. 6. Performance of TIToK w.r.t different values of γ .

4.3.4. Parameter analysis

In order to suppress the noisy activations at the early stage, parameter λ is set to $\lambda = \frac{2}{\exp(-10\theta)} - 1$, where θ changes linearly from 0 to 1 in the training process. While parameter γ is directly set to a fixed value. In this section, we show the performance of TIToK w.r.t. different values of γ from $\{0, 0.05, 0.1, 0.15, 0.2\}$

over 4 adaptation tasks in Fig. 6. From Fig. 6, the performance of TIToK is relatively stable w.r.t. the value of γ over the 4 adaptation tasks. When γ is close to 0, class correlation knowledge is not transferred across domains, when γ becomes large, more attention will be given to class correlation knowledge, and we set $\gamma=0.1$ in our experiments.

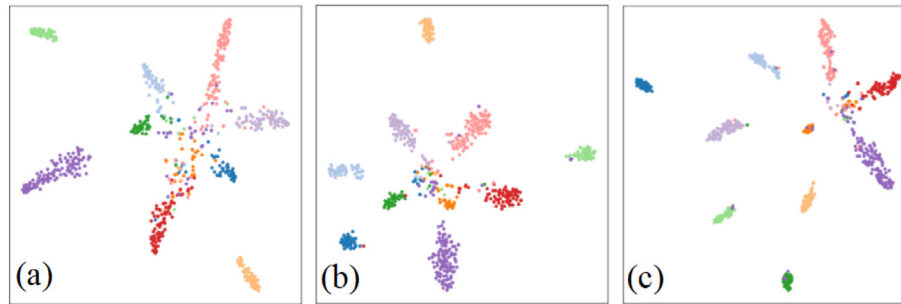


Fig. 7. t -SNE visualization for features learned from (a) DAN, (b) COAL and (c) TIToK, respectively. Different colors represent instances from different classes.

4.3.5. Feature visualization

In this section, we visualize the features learned for target domain over the $R \rightarrow C$ task. The t -SNE (Van Der Maaten, 2014) is adopted for feature visualization, and the results are illustrated in Fig. 7. In Fig. 7(a), under severe class imbalance, it is impossible to predict target instances well with DAN. With learned features from TIToK, target instances are classified more correctly, at the same time, features of each class are clustered more compact, and instances from different classes are better separated compared with those from both DAN and COAL. As a result, through class contrastive transfer, class correlation transfer and discriminative feature alignment, TIToK can achieve encouraging results in bi-imbalanced UDA learning.

5. Conclusion

Data in real-world scenarios are usually imbalanced, including both within-domain and across-domain imbalance in UDA learning. To address this bi-imbalanced issue, we propose an alternative way to transfer the imbalance-tolerant knowledge across domains, and develop a systemic solution TIToK for bi-imbalanced UDA learning. In TIToK, the class contrastive loss is presented and adopted over both source and confident target instances, in order to transfer class contrastive knowledge across domains. Further, the imbalance-invariant correlation knowledge among source classes is transferred in learning to further boost knowledge transfer. Finally, features across domains are aligned by reducing DMMD, which considers both intra-class alignment and inter-class discrimination across domains. Experiments on real benchmark datasets show the competitive of TIToK compared with the state-of-art UDA methods with balanced learning strategies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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